

MATHEMATICS

TARGET : JEE- Advanced

CAPS-17

DEFINITE INTEGRATION

SCQ (Single Correct Type) :

1. The value of the definite integral $\int_2^3 \left[\sqrt{2x - \sqrt{5(4x - 5)}} + \sqrt{2x + \sqrt{5(4x - 5)}} \right] dx$ is equal to
- (A) $4\sqrt{3} - \frac{2\sqrt{2}}{3}$ (B) $4\sqrt{2}$ (C) $4\sqrt{3} - \frac{4}{3}$ (D) $\frac{\sqrt{10}}{2} + \frac{7\sqrt{7} - 5\sqrt{5}}{3\sqrt{2}}$
2. If $\int_{\ln 2}^x \frac{dx}{\sqrt{e^x - 1}} = \frac{\pi}{6}$, then x is equal to
- (A) 4 (B) $\ln 8$ (C) $\ln 4$ (D) none of these
3. The tangent to the graph of the function $y = f(x)$ at the point with abscissa $x = 1$ form an angle of $\pi/3$ and at the point $x = 2$, an angle of $\pi/3$ and at the point $x = 3$, an angle of $\pi/4$. The value of $\int_1^3 f'(x)f''(x)dx + \int_2^3 f''(x)dx$ ($f''(x)$ is supposed to be continuous) is :
- (A) $\frac{4\sqrt{3}-1}{3\sqrt{3}}$ (B) $\frac{3\sqrt{3}-1}{2}$ (C) $\frac{4-\sqrt{3}}{3}$ (D) None of these
4. $\int_{5/2}^5 \frac{\sqrt{(25-x^2)^3}}{x^4} dx$ equals to :
- (A) $\frac{\pi}{3}$ (B) $\frac{2\pi}{3}$ (C) $\frac{\pi}{6}$ (D) none

MCQ (One or more than one correct) :

5. If $\int_{-\alpha}^{\alpha} e^x + \cos x \ln(x\sqrt{1+x^2}) dx > \frac{3}{2}$, then α can be ____.
- (A) 1 (B) 2 (C) 3 (D) 4
6. The value of the definite integral $\int_0^1 \frac{dx}{\sqrt{(x+1)^3}(3x+1)}$ equals ____.
- (A) $\sqrt{2}-1$ (B) $\tan \frac{\pi}{2}$ (C) $\tan \frac{\pi}{8}$ (D) $\tan \frac{5\pi}{12}$
7. If $f(x) = 2^{\{x\}}$, where $\{x\}$ denotes the fractional part of x. Then which of the following is true ?
- (A) f is periodic (B) $\int_0^1 2^{\{x\}} dx = \frac{1}{\ln 2}$
- (C) $\int_0^1 2^{\{x\}} dx = \log_2 e$ (D) $\int_0^{100} 2^{\{x\}} dx = 100 \log_2 e$



Comprehension Type Question:
Comprehension # 1

Consider the integral $I = \int_0^{10\pi} \frac{\cos 6x \cos 7x \cos 8x \cos 9x}{1 + e^{2\sin^3 4x}} dx$

8. If $k \int_0^{\pi/2} \cos 6x \cos 7x \cos 8x \cos 9x dx$ then k is _____.
 (A) 1 (B) 5 (C) 10 (D) 20
9. If $I = \lambda \int_0^{\pi/4} \cos 6x \cos 8x \cos 2x dx$ then λ is _____.
 (A) 5 (B) 10 (C) 20 (D) $\frac{5}{2}$
10. The value of I is _____.
 (A) $\frac{5\pi}{4}$ (B) $\frac{5\pi}{16}$ (C) $\frac{5\pi}{32}$ (D) $\frac{5\pi}{8}$

Numerical based Questions :

11. Given $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$ and $\int_0^1 \frac{\ln x}{1-x^2} dx = -\frac{\pi^2}{\lambda}$ then value of λ is _____.
 12. $\int_1^e \left(\frac{1}{\sqrt{x \ln x}} + \sqrt{\frac{\ln x}{x}} \right) dx = L$ then the value of $\frac{3L}{\sqrt{e}}$ is _____.
 13. If $\int_0^{\frac{\pi}{2}} \sqrt[3]{\sin 2x} \sin x dx = k \int_0^{\frac{\pi}{4}} \sqrt[3]{\cos 2x} \cos x dx$ then the value of k^2 is _____.
 14. $\int_2^4 \frac{3x^2 + 1}{(x^2 - 1)^3} dx = \frac{\lambda}{n^2}$ where $\lambda, n \in \mathbb{N}$ and $\gcd(\lambda, n) = 1$, then find the value of $\lambda + n$
 15. Let $f(x)$ be a function satisfying $f(x) = f\left(\frac{100}{x}\right) \forall x > 0$. If $\int_1^{10} \frac{f(x)}{x} dx = 5$ then find the value of $\int_1^{100} \frac{f(x)}{x} dx$
 16. Evaluate $\frac{2005 \int_0^{1002} \frac{dx}{\sqrt{1002^2 - x^2} + \sqrt{1003^2 - x^2}} + \int_{1002}^{1003} \sqrt{1003^2 - x^2} dx}{\int_0^1 \sqrt{1-x^2} dx} = k$, then find the sum of squares of digits of natural number k .
 17. Let $I_1 = \int_0^{\pi/4} (1 + \tan x)^2 dx$, $I_2 = \int_0^1 \frac{dx}{(1+x)^2(1+x^2)}$ then find the value of $\frac{I_1}{I_2}$

18. If f, g, h be continuous functions on $[0, a]$ such that $f(a-x) = f(x)$, $g(a-x) = -g(x)$ and $3h(x) - 4h(a-x) = 5$, then find the value of $\int_0^a f(x)g(x)h(x)dx$
19. If $f(x) = \frac{\sin x}{x} \forall x \in (0, \pi]$, If $\frac{\pi}{k} \int_0^{\pi/2} f(x)f\left(\frac{\pi}{2}-x\right)dx = \int_0^{\pi} f(x)dx$ then find the value of k .
20. Let a be a real number in the interval $[0, 314]$ such that $\int_{-\pi+a}^{3\pi+a} |x-a-\pi| \sin\left(\frac{x}{2}\right)dx = -16$, then determine number of such values of k .
21. If $f(x) = x + \int_0^1 t(x+t)f(t)dt$, then the value of the definite integral $\int_0^1 f(x)dx$ can be expressed in the form of rational as $\frac{p}{q}$ (where p and q are coprime). Find $(p+q)$.
22. If the minimum of the following function $f(x)$ defined at $0 < x < \frac{\pi}{2}$.

$$f(x) = \int_0^x \frac{d\theta}{\cos \theta} + \int_x^{\frac{\pi}{2}} \frac{d\theta}{\sin \theta} \text{ is equal to } \ln(a+\sqrt{b}) \text{ where } a, b \in \mathbb{N} \text{ and } b \text{ is not a perfect square then find the value of } (a+b)$$

Matrix Match Type :

23. Match the following:

$[x] \rightarrow$ greatest integer less than or equal to x

Column-I	Column-II
(A) $25 \int_0^{\frac{\pi}{4}} (\tan^6(x-[x]) + \tan^4(x-[x]))dx =$	(p) 4
(B) $I_1 = \int_{-2}^2 \frac{x^6 + 3x^5 + 7x^4}{x^4 + 2} dx$ $I_2 = \int_{-3}^1 \frac{2(x+1)^2 + 11(x+1) + 14}{(x+1)^4 + 2} dx$ then $I_1 + I_2 =$	(q) 2
(C) $\frac{2}{\pi} \int_{-1}^3 \left(\tan^{-1} \frac{x}{x^2+1} + \tan^{-1} \frac{x^2+1}{x} \right) dx =$	(r) $\frac{100}{3}$
(D) $f(x) = \frac{2-x}{\pi} \cos \pi(x+3) + \frac{1}{x^2} \sin \pi(x+3)$ where $0 < x < 4$. The number of points of local extrema are	(s) 5
	(t) $\frac{102}{3}$

(A) $A \rightarrow s; B \rightarrow r; C \rightarrow q; D \rightarrow q$

(B) $A \rightarrow s; B \rightarrow s; C \rightarrow q; D \rightarrow r$

(C) $A \rightarrow s; B \rightarrow r; C \rightarrow s; D \rightarrow q$

(D) $A \rightarrow s; B \rightarrow r; C \rightarrow r; D \rightarrow q$

Subjective Based Questions :

24. Evaluate $\int_0^1 \frac{1}{(5+2x-2x^2)(1+e^{(2-4x)})} dx$

25. Evaluate $\int_0^1 (\{2x\} - 1)(\{3x\} - 1) dx$, where $\{ \}$ denotes fractional part of x .

ANSWER KEY OF CAPS-17

1. (D)	2. (C)	3. (D)	4. (A)	5. (ABCD)
6. (AC)	7. (ABCD)	8. (C)	9. (B)	10. (D)
11. (8)	12. (6)	13. (2)	14. (61)	15. (10)
16. (29)	17. (4)	18. (0)	19. (2)	20. (25)
21. (65)	22. (11)	23. (A)	24. $\frac{1}{\sqrt{11}} \ln \frac{\sqrt{11}+1}{\sqrt{11}-1}$	25. $(\frac{19}{72})$